

CSIS 2430 Project Plan

Course/Term: CSIS 2430

Project: Project 1

Team: Helen Le, Owen Francy, Mustafa Alrubaye

Start Date: Sep 13th

Due Date: October 4th 2026

Checkpoint/mid-project Date: Sep 20th

Team & Roles

Team Member	Role	Primary Responsibilities	Availability/notes
Owen	Implementation Lead	owns core implementation, integration, and main branch health.	Mon - Thurs: 12 pm - basically whenever Fri: 9 am - 2 pm Sat: 9 am - 12:30 pm and ~10:30 pm - 1 Sun: 8 pm - 12 am
Mustafa	Verification Lead	owns test plan, edge cases, and verification evidence.	Monday 3:30 - 11 Tuesday 1 - 5 and 7:30 - 11 Wednesday 3:30 - 5 and 10-11 Thursday 1 - 5 and 7:30 - 11 Friday 3:30 - 11 Saturday 3:30 to whenever Sunday 3:30 - 11
Helen	Communications Lead	owns report assembly, run instructions, and deliverable packaging.	(work hours not accounted for) Mon, Wed, Thurs: 12:30 pm - whenever Tues: 12:30 pm - 4:30 pm Fri-Sun: whenever

Repository and Access

Platform: GitHub

Repository URL: <https://github.com/OwenFrancy/CS2430-Project1.git>

Instructor invited as viewer? Not Yet, will count as final submission as well

Repo Structure Requirement (summary): Include /docs with pseudocode, UML/diagram, design notes; keep updated at checkpoint.

Project Deadlines

Due	Assignment
Sep 13	Weekly Status Report Week 1
Sep 20	Weekly Status Report Week 1
Sep 20	Checkpoint
Sep 27	Weekly Status Report Week 3
Oct 4	Peer Review
Oct 4	Algorithm Performance - Project Due

Milestones (Timeline)

Target date	Milestone	Definition of Done	Status	Evidence Pointer
Sep 20	Repo set up + running skeleton	Repo is made & there is a runnable skeleton	D	sorting
Sep 24	Core implementation	Required features implemented; compiles	IP	
Sep 28	Verification (testing)	Edge cases are covered; document actual vs expected results	NS	
Oct 1	Report/Video drafts	Report draft done; video outline/script notes	NS	
Oct 4	Final integration + submission	All deliverables ready	NS	

Task Backlog (Assignments + Status)

ID	Task	Owner	Due Date	Status	Evidence Link/Pointer	Notes/Dependencies
1	Create Repo (contains runnable skeleton + current implementation branch)	Owen	9/20	D	sorting	
2	UML diagram	Owen	9/20	D	docs/Project1 UML Diagrams.png	
3	Design Notes/Decisions	Helen	9/20	D	docs/DesignNotes	
4	/docs folder with Project Plan and 2 design artifacts	Team	9/20	D	docs/...	
5	Verification	Mustafa				
6	Team report	Helen	10/3			
7	Team screencast video (<=10mins)	Helen	10/3			
8	Team code package.zip containing: all src code, README, CONTRIBUTIONS.md, data output files	Helen	10/3			

Verification Plan (Verification Lead)

1. Correctness check
 - Verify that it generates/preduces 24 unique ordered arrangements when the input is 4. This would also be done different unique arrays without them duplicating.
 - Verify that the different sorts (mergesort, quicksort, etc.) return fully sorted and correctly sorted arrays for every generated combination.
2. Edge cases to verify
 - Test how the sorting algorithm would handle the smaller arrays (for example, $n = 1$, $n = 2$) to see how it would run.
 - Ensure it doesn't break with different inputs
3. Expected vs. actual outputs
 - Check if it actually outputs the 10 best and 10 worst outputs, along with the overall average
4. Performance checks
 - Make sure the program starts from zero before trying to run it again.
 - Make sure that the algorithm is counting the exact same way for all 4 algorithms so the test is not unfair in some areas.

Risks/Blockers

Risk/Blocker	Impact	Mitigation Plan	Owner	Status

Update Log

Date	Update Summary	Blockers/Resolution	Owner	Evidence Pointers
9/20	Repo created		Owen	
9/20	UML Diagram made		Owen	docs/Project1 UML Diagrams.png
9/20	Design notes		Helen	docs/DesignNotes